The Wang-Landau algorithm in general state spaces: Applications and convergence analysis

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Abstract

The Wang-Landau algorithm (Wang and Landau (2001)) is a recent Monte Carlo method that has generated much interest in the Physics literature due to some spectacular simulation performances. The objective of this paper is two-fold. First, we show that the algorithm can be naturally extended to more general state spaces and used to improve on Markov Chain Monte Carlo schemes of more interest in Statistics. In particular, we detail with an example how the method can be used to improve on trans-dimensional MCMC samplers. In a second part, we study the asymptotic of the algorithm. We show that with an appropriate choice of the step-size, the algorithm is consistent and a strong law of large numbers hold under some fairly mild conditions. To the best of our knowledge, this gives the first rigorous proof of the convergence of the Wang-Landau algorithm.

Key words: Monte Carlo Methods, Muticanonical sampling, Wang-Landau algorithm, Trans-dimensional MCMC, Adaptive MCMC, Stochastic approximations, Geometric ergodicity

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1 Introduction

Monte Carlo methods have been around for more than a century. The field has been revitalized by physicists during the second World War and N. Metropolis coined the name "Monte Carlo". Ever since, physicists have been at the forefront of the methodological research in the field. One of their latest addition is an algorithm proposed by F. Wang and D. P. Landau (Wang and Landau (2001)). The Wang-Landau (WL) algorithm has been successfully applied to some complex sampling problems in physics (see e.g. Lee et al. (2005) and the references therein). The algorithm is a natural extension of *multicanonical sampling*, a method due to B. A. Berg and T.

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Neuhaus (Berg and Neuhaus (1992)). Briefly, if π is the probability measure of interest, the idea behind multicanonical sampling is to obtain an importance sampling distribution by partitioning the state space along the energy function ($-\mathrm{Log}\ \pi(x)$) and reweighting appropriately π in each component of the partition so as to balance all the components. The method is often criticized for the difficulty involved in computing the weights. The main contribution of the WL algorithm is in proposing an efficient algorithm that *simultaneously* computes the balancing weights and samples from the reweighted distribution.

The objective of this paper is to take a more probabilitic look at the WL algorithm and to explore its potential for Monte Carlo simulation problems of more direct interest to statisticians. We achieve this goal by proposing an abstract, general state space version of the algorithm. Then we show that the WL algorithm offers an effective strategy to improve on trans-dimensional MCMC. Trans-dimensional MCMC is a set of specialized MCMC algorithms developed mainly for Bayesian modeling with model incertainty. The seminal paper on the topic is Green (1995). The same idea of reweighting advocated by the WL algorithm can also be introduced in this context. Call $\pi(x_k, k)$ the posterior distribution of a Bayesian model with model incertainty. Marginalizing over the parameter x_k gives the posterior distribution of the model $\pi(k)$. In this framework, the WL strategy is as follows. Instead of sampling from π, we build another posterior distribution $\pi^*(x_k, k)$ by reweighting appropriately each model in such a way that, under π^*, the posterior distribution of the models is $\pi^*(k) \equiv c$. Plainly, under π^*, all the models have the same weight and will be allocated the same (MCMC) computing resource. The question is naturally asked whether this is a sensible strategy? By analyzing some finite states space approximating chains, we show that under certain conditions, the reweighting strategy yields a Markov chain with improved mixing. This point is further illustrated with an example from Bayesian normal mixture modeling.

From a probabilitic standpoint, the WL algorithm is an interesting example of adaptive Markov Chain Monte Carlo (MCMC). Adaptive MCMC is an approach to Monte Carlo simulation where the transition kernel of the algorithm is sequentially adjusted over time in order to achieve some prescribed optimality. Some early work on the subject includes Gilks et al. (1998), Andrieu and Robert (2001), Haario et al. (2001). See also Brockwell and Kadane (2005), Mira and Sargent (2003). Early theoretical analysis includes (Atchade and Rosenthal (2005), Andrieu and Moulines (2006), Andrieu and Atchade (2005), Rosenthal and Roberts (2005)). We take a similar pathwise approach to analyze the WL algorithm. The analysis of the WL algorithm is not a straightforward application of the theory in the above mentioned papers because of the delicate stochastic control of the step-size. The key word is *stability*. We say that the WL algorithm is *stable* if no component of the partition receives infinitely more visits than any other component as $n \to \infty$. On a *stable* sample path and under appropriate conditions, we show that the WL algorithm learns the optimal weights and satisfies a strong law of large numbers (Theorem 5.1). In the case where the kernels of the algorithm are Metropolis-Hastings kernels, we show that these conditions simplify considerably and boil down to checking the geometric ergodicity of simple Metropolis-Hastings kernels. In the specific context of multicanonical sampling with Independent-Metropolis kernels (which includes the original the WL algorithm), we show that the algorithm is *stable* and that the above mentioned limit results hold. In general, the *stability* of the algorithm turns out to be difficult to check. But we obtain a result (Theorem 5.2) which guarantees *stability* in a number of cases (including cases where the states space is discrete or compact).

It came to our attention after the first draft of this paper that a similar generalization of the WL algorithm has been proposed independently by F. Liang and coworkers (see e.g. Liang et al. (2007)). The theory developed here applies to their algorithm as well.

Our proposed generalization to the WL algorithm is presented in Section 2. Some particular cases are discussed in Section 4 which comprises the application to trans-dimensional MCMC in Section 3 with an example from Bayesian mixture modeling. The theoretical analysis is discussed in Section 5 but the proofs are postponed to Section 7 to facilitate the flow of ideas.

2 The Wang-Landau algorithm

In multicanonical sampling, we are given a state space $\mathcal{X}$ and a probability measure π. $\mathcal{X}$ is then partitioned, $\mathcal{X} = \cup \mathcal{X}_i$ and π is reweighted in each component $\mathcal{X}_i$. An abstract way to do the same and much more is the following. We start with $(\mathcal{X}_i, \mathcal{B}_i, \lambda_i)$ $i = 1, \ldots, d$, a finite family of measure spaces where λ_i is a σ-finite measure. We introduce the union space $\mathcal{X} = \bigcup_{i=1}^d \mathcal{X}_i \times \{i\}$. We equip $\mathcal{X}$ with the $\sigma-$algebra $\mathcal{B}$ generated by $\{(A_i, i),\ i \in \{1, \ldots, d\},\ A_i \in \mathcal{B}_i\}$ and the measure λ satisfying $\lambda(A, i) = \lambda_i(A)\mathbf{1}_{\mathcal{B}_i}(A_i)$. Let $h_i :\ \mathcal{X}_i \to \mathbb{R}$ be a nonegative measurable function and define $\theta^*(i) = \int_{\mathcal{X}_i} h_i(x)\lambda_i(dx)/c$ where $c = \sum_{i=1}^d \int_{\mathcal{X}_i} h_i(x)\lambda_i(dx)$. We assume that $\theta^*(i) > 0$ for all $i = 1, \ldots, d$. Consider the following probability measure on $(\mathcal{X}, \mathcal{B})$:

$$\pi^*(dx, i) \propto \frac{h_i(x)}{\theta^*(i)} \mathbf{1}_{\mathcal{X}_i}(x)\lambda_i(dx). \tag{1}$$

Our objective is to sample from π^*. The problem of sampling from such distribution arises in a number of different contexts. For example, if π is a probability measure of interest on some space $(\mathcal{X}, \mathcal{B})$, we can partition $\mathcal{X}$ with a partition $(\mathcal{X}_i)$ built along the energy function $-\log(\pi)$ and reweight π by $\pi(\mathcal{X}_i)$ in $\mathcal{X}_i$. The sampling problem then becomes of the form (1). This powerful strategy appeared first in the Physics literature as *multicanonical sampling* (Berg and Neuhaus (1992)). This is discussed in more detail is Section 4.1.

Sampling from (1) also arises naturally when optimizing the *simulated tempering* algorithm (Marinari and Parisi (1992), Geyer and Thompson (1995)). In simulated tempering, the states space $\mathcal{X}$ is not partitioned but instead some auxiliary distributions $\pi_2, \ldots, \pi_d$ are introduced (take $\pi_1 = \pi$). These distributions are chosen close to π but easier to sample from. For good performances, one can impose that all the distributions have the same weight. Taking each probability space $(\mathcal{X}, \mathcal{B}, \pi_i)$ as a component in the formalism above leads to a sampling problem of the form (1). Multicanonical sampling and simulated tempering have been combined in Atchade and Liu (2006) giving an algorithm which can also be framed as (1).

One of the objectives of this paper is to show that sampling from (1) can also be an efficient strategy to improve on MCMC samplers for Bayesian inference with model incertainty (the so-called trans-dimensional MCMC). This is detailed in Section 3.

The main obtacle in sampling from π^* is that the normalizing constants θ^* are not known. A trial-and-error solution is often adopted. A number of adaptive, automatic solutions, including stochastic approximation algorithms have been also been proposed in the literature with mitigated successes (see e.g. Geyer and Thompson (1995)). The contribution of Wang and Landau

(2001) is to introduce an efficient algorithm that estimates θ^* and samples from π^* simultaneously in a discrete state space setting. To discuss the WL algorithm in our general framework, we introduce the family of probability measures $\{\pi_\theta,\ \theta \in \mathbb{R}^d,\ \theta(i) > 0\}$ on $(\mathcal{X}, \mathcal{B})$ defined by:

$$\pi_\theta(dx, i) \propto \frac{h_i(x)}{\theta(i)} \mathbf{1}_{\mathcal{X}_i}(x) \lambda_i(dx). \tag{2}$$

We assume that for all $\theta \in \mathbb{R}^d$, $\theta(i) > 0$, we have at our disposal a transition kernel P_θ on $(\mathcal{X}, \mathcal{B})$ with invariant distribution π_θ. How to build such Markov chain P_θ typically depends on the particular instance of the algorithm. For example, when sampling from $h_i \mathbf{1}_{\mathcal{X}_i}$ is possible, we typically iterate through x and i with a Gibbs or a Metropolis-winthin-Gibbs scheme. Otherwise a general Metropolis-Hastings scheme on the joint (x, i) might be performed. More details are given below.

The WL algorithm is an adaptive MCMC algorithm where θ^* is estimated during the simulation. At time n, suppose that we have the sequence $(X_0, I_0), \ldots, (X_n, I_n)$ and some estimate θ_n of θ^*. We obtain (X_{n+1}, I_{i+1}) by sampling from $P_{\theta_n}(X_n, I_n; \cdot)$. Then $(X_0, I_0, \ldots, X_{n+1}, I_{n+1})$ is used to obtain a new estimate θ_{n+1} of θ^*. The power of the method comes from the fact that it builds the estimate of θ^* on a multiplicative scale with a fast early convergence toward θ^*:

$$\theta_n(i) = \frac{e^{-\gamma \sum_{k=1}^n \mathbf{1}_{\{I_k = i\}}}}{\sum_{j=1}^d e^{-\gamma \sum_{k=1}^n \mathbf{1}_{\{I_k = j\}}}}.$$

The step-size γ is gradually lowered during the simulation as θ_n gets closer to θ^*. This is done carefully to avoid a premature convergence of the step-size a well known problem in the stochastic approximation literature.

For the details, let $\{\gamma_n\}$ be a sequence of positive numbers, $\kappa_0 = 0$, $a_0 = 0$, $c \in (0, 1)$, $\phi_0 \in \mathbb{R}^d$ such that $\phi_0(i) > 0$ for all i be given. The initial estimate of θ^* is $\theta_0(i) = \phi_0(i) / \sum_j \phi_0(j)$. For $n \geq 0$ and $i \in \{1, \ldots, d\}$, let $v_{n,k}(i)$ be the proportion of visits to $\mathcal{X}_i \times \{i\}$ between times $n+1$ and k. That is, $v_{n,n}(i) = 0$ and for $k \geq n+1$, let $v_{n,k}(i) = \frac{1}{k-n} \sum_{j=n+1}^k \mathbf{1}\{I_j = i\}$. At time $n \geq 1$, given $\mathcal{F}_n = \sigma(X_0, I_0, \ldots, X_n, I_n)$, if:

$$\max_i \left| v_{\kappa_{a_{n-1}}, n}(i) - \frac{1}{d} \right| \leq \frac{c}{d}, \tag{3}$$

we set $a_n = a_{n-1} + 1$, and $\kappa_{a_n} = n$. Otherwise, we set $a_n = a_{n-1}$. Then, with $\theta_n(i) = \phi_n(i) / \sum_j \phi_n(j)$ we sample $(X_{n+1}, I_{n+1}) \sim P_{\theta_n}(X_n, I_n; \cdot)$ and set $\phi_{n+1}(i) = \phi_n(i)(1 + \gamma_{a_n} \mathbf{1}_{\{I_{n+1} = i\}})$.

The number a_{n-1} determines the element of the sequence $\{\gamma_n\}$ to be used at time n and also represents the number of step-size changes before time n. As for $\kappa_{a_{n-1}}$, it represents the time of the last step-size modification before n. After time $\kappa_{a_{n-1}}$, the algorithm monitors the visits to each component until all the proportions of received visits come within c/d of $1/d$. The algorithm is most effective when c is chosen within the range $0.01 - 0.20$ depending on the complexity of the simulation. Small values of c being better for large simulation problems.

Algorithm 2.1. *The Wang-Landau algorithm*

At some time $n \geq 0$, given X_n, I_n, ϕ_n, θ_n a_n and κ_{a_n}:

(i) *Sample $(X_{n+1}, I_{n+1}) \sim P_{\theta_n}(X_n, I_n; \cdot)$.*

(ii) *For $i = 1, \ldots, d$, set $\phi_{n+1}(i) = \phi_n(i)\left(1 + \gamma_{a_n} \mathbf{1}_{\{I_{n+1}=i\}}\right)$ and $\theta_{n+1}(i) = \phi_{n+1}(i)/\sum_j \phi_{n+1}(j)$.*

(iii) *If $\max_i \left|v_{\kappa_{a_n}, n+1}(i) - \frac{1}{d}\right| \leq c/d$ then set $a_{n+1} = a_n + 1$ and $\kappa_{a_{n+1}} = n + 1$. Otherwise, we set $a_{n+1} = a_n$.*

We will say that the algorithm is *stable* if no component receives excessively more visits than any other component.

Definition 2.1. The Wang-Landau algorithm is said to be stable if

$$\max_{i,j} \limsup_{n\to\infty} n(v_n(i) - v_n(j)) < \infty, \ a.s. \tag{4}$$

This is an essential property of the algorithm. When the algorithm is *stable*, all the stopping times κ_l defined above are finite and the step-size γ_{a_n} will gradually converges to 0. Moreover our key result (Theorem 5.1 below) states that on a path where the algorithm is *stable*, $\theta_n \to \theta^*$ and a strong law of large numbers hold.

3 Application to Trans-dimensional MCMC

It is often the case in Statistics that many alternative models are considered for the same data. One is then interested in issues like model comparison, model selection and averaging. Let

$f(Data|k, x_k)$ be the likelihood of model k with parameter x_k. Assume that we have a finite number d of models and that $x_k \in (\mathcal{X}_k, \mathcal{B}_k, \lambda_k)$. Let $\mathcal{X} = \bigcup_{i=1}^{d} \mathcal{X}_i \times \{i\}$ be the union space equipped as above with the σ-algebra $\mathcal{B}$ and the σ-finite measure λ. $(\mathcal{X}, \mathcal{B}, \lambda)$ is the natural space to consider when dealing both with model incertainty and parameter estimation. In the Bayesian framework, a prior density (wth respect to λ) $p(x_k, k)$ in $(\mathcal{X}, \mathcal{B})$ is specified for (x_k, k). The posterior distribution of (x_k, k) is therefore $\pi(x_k, k) \propto h_k(x_k) = f(Data|k, x_k)p(x_k, k)$. Trans-dimensional MCMC is a set of specialized MCMC algorithms to sample from distributions like π defined on spaces of variable dimensions. The reversible-jump algorithm of Green (Green (1995)) is the most popular such sampler.

Understanding the geometry of union spaces and designing efficient trans-dimensional moves is the subject of current research in MCMC (Brooks et al. (2003)). As currently available, these samplers typically have a poor mixing, specifically slowed down by the trans-dimensional moves. In these cases and unsurprisingly, quantities like Bayes factors tend to be poorly estimated. In the framework above, the Bayes factor of model i to model j is $B_{ij} := \theta^*(i)p(j)/(\theta^*(j)p(i))$, where $\theta^*(i) \propto \int_{\mathcal{X}_i} \pi(x_i, i)\lambda_i(dx_i)$ and $p(i) = \int p(x_i, i)\lambda_i(dx_i)$.

In the spirit of the WL algorithm, an alternative to sampling from π is to sample from the distribution

$$\pi^*(dx_i, i) \propto \frac{h_i(x_i)}{\theta^*(i)} \mathbf{1}_{\mathcal{X}_i}(x)\lambda_i(dx_i). \tag{5}$$

By such re-weighting, we give the same posterior weight to all the models. The WL algorithm then offers an effective strategy to sample from π^* and we recover π by importance sampling.

Whether this reweighting strategy is a good strategy here is clearly problem specific. If the main objective of the model is to find the best model and estimate its parameter then reweighting might not be such a good idea as it clearly diverts computing resource away from the most probable model. But in most cases, one is more interested in comparing and/or averaging the models. In these cases reweighting seems more attractive. Beyond these model-specific arguments, we argue that reweighting can improve on the mixing of the algorithm. We can make this statement precise in the following crude way.

Denote by P (resp. P^*) the trans-dimensional transition kernel to sample from π (resp. π^*).

We assume that P and P^* are Metropolis-Hastings kernels with the same proposal kernel Q. In practice, the proposal kernel Q is given. It is then transformed through the Metropolis-Hastings map to target either π or π^*. Under π (resp. π^*), the marginal stationary distribution on $\{1,\ldots,d\}$ is $(\theta^*(i))$ (resp. the uniform distribution). Thus we can approximate P by a $d \times d$ Metropolis-Hastings transition matrix $\bar{P}$ with invariant distribution $(\theta^*(i))$ and proposal q_{ij} given by:

$$q_{ij} = \frac{1}{\theta^*(i)} \int_{\mathcal{X}_i} \pi(dx,i) Q\left(x,i;\mathcal{X}_j \times \{j\}\right).$$

So q_{ij} is the average probability of proposing a jump from i to j in stationarity. Similarly we approximate P^* by the Metropolis-Hastings transition matrix $\bar{P}^*$ with uniform invariant distribution on $\{1,\ldots,d\}$ and same proposal q_{ij}. We have:

$$\bar{P}(i,j) = \begin{cases} \min\left(1, \frac{\theta^*(j)}{\theta^*(i)} \frac{q_{ji}}{q_{ij}}\right) q_{ij} & \text{if } i \neq j, \\ 1 - \sum_{j \neq i} \bar{P}(i,j) & \text{otherwise.} \end{cases}$$

and

$$\bar{P}^*(i,j) = \begin{cases} \min\left(1, \frac{q_{ji}}{q_{ij}}\right) q_{ij} & \text{if } i \neq j, \\ 1 - \sum_{j \neq i} \bar{P}^*(i,j) & \text{otherwise.} \end{cases}$$

The following result is then immediat.

Proposition 3.1. *If q is symmetric ($q_{ij} = q_{ji}$) then $\bar{P}^*$ has a perfect mixing while $\bar{P}$ can mix arbitrarily slowly depending on θ^*.*

Remark 3.1.

1. Although based on an approximate argument, this result suggests that the reweighting strategy will improve on mixing if the proposal mechanism is not too skewed. In practice, proposals are often based on symmetric random walks and in these cases, reweighting should help.

2. One important limitation of this analysis, is the fact that the approximating chains only capture the transitions from one model to another and the mixing problems thereof. But the overall performance of the reweighting strategy will also depends on the mixings of the within-model samplers (P and P^* restricted to each component $(\mathcal{X}_i, \mathcal{B}_i, \lambda_i)$). If all the within-model samplers have about the same mixing, our analysis above carries over.

Otherwise, it is intuitively clear that the components with the worst mixings should be given more weight. Therefore targeting a uniform distribution over the models will not be a good strategy. More work need to be done on this problem.

3.1 A simulation example: Bayesian mixture modeling

We present a simulation example to illustrate the idea. In this example we show that the WL algorithm can improve on mixing time and Bayes factor estimation in trans-dimensional MCMC. The example is taken from (Richardson and Green (1997)).

Let $y = (y_1, \ldots, y_n)$ be a vector of n univariate observations. We assume that $(y_1, \ldots, y_n)$ are i.i.d. realisations from a mixture of Gaussian univariate distribution and we would like to infer the weights p_i, the means μ_i and variances σ_i^2 of the components of the mixture. The number of components in the mixture is unknown and also need to be estimated. For a k-component mixture, the parameter vector is $x_k = (p_1, \ldots, p_k, \mu_1, \ldots, \mu_k, \sigma_1, \ldots, \sigma_k, \beta) \in \mathcal{X}_k = (0, \infty)^k \times \mathbb{R}^k \times (0, \infty)^k \times (0, \infty)$, where β is a parameter described below. Define $\mathcal{X} = \bigcup_{k=1}^{d} \mathcal{X}_k \times \{k\}$. The likelihood of the model given the observation y is:

$$L(x_k, k|y) = \prod_{j=1}^{n} \sum_{i=1}^{k} \omega_i \mathcal{N}(y_j; \mu_i, \sigma_i^2),$$

where $\omega_i = p_i / \sum p_j$, $\mathcal{N}(\cdot, \mu, \sigma^2)$ is the density of the normal distribution with mean μ and variance σ^2. With a prior distribution $f(x_k, k)$ on (x_k, k) the posterior distribution of interest is:

$$\pi(x_k, k) \propto f(x_k, k) \prod_{j=1}^{n} \sum_{i=1}^{k} \omega_i \mathcal{N}(y_j; \mu_i, \sigma_i^2).$$

We assume the following prior distribution decomposition:

$$\begin{aligned} f(x_k, k) &= f(k) f(x_k|k) \\ &= f(k) \prod_{i=1}^{k} f(p_i) f(\mu_i) f(\sigma_i|\beta) f(\beta). \end{aligned}$$

More specifically, for the priors, we assume that $k \sim \mathcal{U}\{1, \ldots, d\}$ the uniform distribution on $\{1, \ldots, d\}$; that $p_i \sim G(\delta, 1)$, a Gamma distribution with parameter δ and 1; $\mu_i \sim N(\xi, \kappa^{-1})$, the normal distribution with parameters ξ, κ^{-1}; $\sigma_i^{-2}|\beta \sim G(\alpha, \beta)$ and $\beta \sim G(g, h)$. The hyperparameters ξ, κ, α, g and h are estimated from the data as in Richardson and Green (1997).

Given the posterior distribution π, we implement a trans-dimensional move using the split-merge reversible jump of Richardson and Green (1997). We sample from the fixed dimension distribution using a Metropolis-Hastings algorithm without the allocation auxiliary variable (see e.g. Marin and Mengersen (2005)). We compare this sampler with a similar sampler that also implements the WL algorithm. Both samplers have about the same computational costs. We compare the two samplers in terms of acceptance rate and on how well they estimate the Bayes factors. We estimate the acceptance rate of each sampler on the three dataset considered in Richardson and Green (1997). For the simulation, each sampler is run independently for $100,000$ iterations starting from the same point. For the WL recursion, we use $\gamma = 30$ and in order to check the flatness of the histogram, we choose $c = 0.01$. As we can see in Table 2, the results indicate a more improved between-model acceptance rate. We want to stress here that the WL algorithm has a very low computational cost. In the simulations described next, we also find that the WL reversible jump leads to a better estimation of the Bayes factors.

Dataset	WL reversible jump	Plain reversible jump
Galaxy	24	5
Enzyme	28	6
Acidity	20	9

Table2: Between-model acceptance rates (in percentage) on the normal mixture problem. Based on $100,000$ iterations. Both samplers have the same computational costs.

To compare the samplers on Bayes factor estimation, we generate an artificial dataset of $n = 300$ independent sample points from a mixture of $k = 3$ components with weights $\omega = (0.1, 0.2, 0.7)$, with means $\mu = (-5.0, 0.0, 4.0)$ and variances $\sigma^2 = (1, 0.01, 0.09)$. We compare 30 competing models with mixture components from 1 to 30. We estimate the posterior distribution of each model using both samplers based on 100, 000 iterations. In order to compare the estimates, we repeat the simulation 100 times. The results are reported in Table 3 only for models with number of mixture components from 3 to 7 which account for about 90% of the posterior distribution. We reports the average posterior distribution over the 100 replications together with the mean square error given that we know the true model. Once again, a significant gain can be achieved by combining the plain RJ with the re-weighting approach advocated here.

		3	4	5	6	7
Average post. prob.	WL-RJ	0.628	0.210	0.095	0.005	0.007
	Plain RJ	0.150	0.326	0.266	0.155	0.066
Mean Square Error	WL-RJ	0.559	0.373	0.252	0.192	0.055
	Plain RJ	0.905	0.493	0.408	0.276	0.146
	Improvement(%)	61	32	62	44	165

Table2: Models Posterior distribution: average and mean square error over 100 replications.

4 Other special cases

In this section, we detail briefly some applications of the general algorithm to multicanonical sampling, simulated tempering and transdimensional MCMC.

4.1 Multicanonical sampling

The multicanonical sampling algorithm is a powerful algorithm proposed by Berg and Neuhaus (1992). It holds the potential of improving on mixing times of classical MCMC algorithms. It fits naturally in the framework above. But the implementation can be tedious. Assume that we want to sample from a probability measure $\pi(dx) \propto h(x)\lambda(dx)$ on some probability space $(\Sigma, \mathcal{A}, \lambda)$. We use the energy function $E(x) = -\log(h(x))$ to build a d-component partition $(\mathcal{X}_i)_i$ of Σ. $\mathcal{X}_i = \{x \in \Sigma :\ E_{i-1} < E(x) \leq E_i\}$, where $-\infty \leq E_0 < E_1 < \ldots < E_d \leq \infty$ are predefined values. Denote $\theta^*(i) = \pi(\mathcal{X}_i)$ and assume $\theta^*(i) > 0$. As above, we introduce the union space

$\mathcal{X} = \bigcup \mathcal{X}_i \times \{i\}$. The idea of multicanonical sampling is to sample from π^* given by:

$$\pi^*(dx, i) \propto \frac{h(x)}{\theta^*(i)} \mathbf{1}_{\mathcal{X}_i}(x)\lambda(dx),$$

which is of the form (1). There is a simpler formulation of the algorithm. Since the component of the partition to which a point x belongs can be obtained from x itself, multicanonical sampling is equivalent to sampling from π^* on $(\Sigma, \mathcal{A})$ given by:

$$\pi^*(dx) \propto \sum_{i=1}^{d} \frac{h(x)}{\theta^*(i)} \mathbf{1}_{\mathcal{X}_i}(x)\lambda(dx), \tag{6}$$

and the union space formalism is not needed. After sampling from π^* in (6), a straightforward importance sampling estimate allows to recover π. The algorithm tries to break the barriers in the energy landscape of the distribution by reweighting each component $\mathcal{X}_i$. Clearly, the success depends heavily on a good choice of the energy rings $E_0, \ldots, E_d$. This typically requires some prior information on π or some pilot simulations. Another subtle problem of the method is how to design the sampling kernel P_θ with invariant distribution $\pi_\theta(dx) \propto \sum_{i=1}^{d}(h(x)/\theta(i))\mathbf{1}_{\mathcal{X}_i}(x)\lambda(dx)$, $\theta \in \mathbb{R}^d$, $\theta(i) > 0$. It is important that such sampler be able to propose jumps from one component to another.

Taking Σ as a discrete space and $\mathcal{X}_e = \{x \in \Sigma : \ E(x) = e\}$, $e \in \{e \in \mathbb{R} : \ E(x) = e \text{ for some } x \in \Sigma\}$ in the above formulation yields the original WL algorithm (Wang and Landau (2001)).

4.2 Simulated Tempering

The method can be applied to the simulated tempering of Marinari and Parisi (1992) and Geyer and Thompson (1995) by taking $(\mathcal{X}_i, \mathcal{B}_i, \lambda_i) \equiv (\mathcal{X}_1, \mathcal{B}_1, \lambda_1)$ and $h_i = h^{1/t_i}$, $1 = t_1 < t_2 < \ldots < t_d$. Simulated tempering is a well known Monte Carlo strategy. Assume that the distribution of interest is $\pi_1(dx) \propto h(x)\lambda(dx)$. Typically for large temperature t, $h^{1/t}$ is a more well-behaved distribution for which faster mixing Markov chains can be built. In simulated tempering, we try to take advantage of these faster mixing chains, by targeting the distribution:

$$\pi_\theta(dx, i) = \frac{h^{1/t_i}(x)}{\theta(i)} \mathbf{1}_{\mathcal{X}_i}(x)\lambda_1(dx), \tag{7}$$

on the union space $(\mathcal{X}, \mathcal{B}, \lambda)$. A MCMC sampler P_θ with invariant distribution π_θ is readily designed. Typically, P_θ takes the form $P_\theta = B_\theta W$ where W moves x while the temperature is fixed and B_θ moves the temperature while x is fixed. We refer the reader to Geyer and Thompson (1995) for more details. By standard importance sampling techniques, we can convert samples from the higher temperature distributions to also estimate π_1. The method holds for any $\theta \in \mathbb{R}^d$, $\theta(i) > 0$. But the choice of θ can significantly impact the efficiency. Heuristically, it seems that, to improve on mixing, we need a θ that allows to sample from fast converging distributions (but close to π); but since π_1 is the distribution of interest, for statistical efficiency, we need a θ that favors π_1. One easy way to resolve this trade-off is to choose θ such that all the distributions are equally visited. For this we need to choose $\theta(i) = \theta^*(i) \propto \int h_i(x)\lambda_1(dx)$ and sample from

$$\pi^*(dx, i) \propto \frac{h^{1/t_i}(x)}{\theta^*(i)} \mathbf{1}_{\mathcal{X}_i}(x)\lambda_1(dx),$$

which can be done with the WL algorithm.

5 Some theoretical results

We look at some theoretical aspects of the algorithm. We investigate the convergence of θ_n to θ^* and a strong law of large numbers. We also derive some conditions under which the algorithm is *stable*. To maintain the flow of ideas, some of the proofs are postponed to Section 7.

5.1 Ergodicity

Let $\Theta = \{\theta \in \mathbb{R}^d : \sum_{i=1}^d \theta(i) = 1,\ \theta(i) \in (0,1),\ i = 1, \ldots, d\}$ and $((X_0, I_0), \theta_0) \in \mathcal{X} \times \Theta$ be the initial state of the algorithm. This initial state will be considered fixed but arbitrary. Let $\{\gamma_n\}$ be the step-size sequence. Let Pr be the distribution of the process $\{X_n, I_n, \theta_n\}$ started at (X_0, I_0, θ_0) with step-size sequence $\{\gamma_n\}$ and denote $\mathbb{E}$ the expectation with respect to Pr. To simplify the notations, we will omit to make explicit the dependence of Pr on (X_0, I_0, θ_0) and $\{\gamma_n\}$. All statements made almost surely will be with respect to Pr. For $\theta \in \mathbb{R}^d$, $|\theta|$ denotes the Euclidean norm of θ. For any $\varepsilon \in (0, \varepsilon_*)$, where $\varepsilon_* = \min_i \theta^*(i)$, define $\Theta_\varepsilon = \{\theta \in \Theta,\ \theta(i) \geq \varepsilon,\ i = 1, \ldots, d\}$.

Our main assumption asserts that the family $\{P_\theta,\ \theta \in \Theta_\varepsilon\}$ is Lipschitz and uniformly V-ergodic. See e.g. Andrieu and Moulines (2006) for some examples of MCMC samplers where these assumptions hold.

(**A1**) We assume the existence of a measurable function $V:\ \mathcal{X} \longrightarrow [1,\infty)$, a set $C \subset \mathcal{X}$, a probability measure ν on $(\mathcal{X},\mathcal{B})$ such that $\nu(C) > 0$ with the following property. For all $\varepsilon \in (0,\varepsilon_*)$ we can find constants $\lambda_\varepsilon \in (0,1)$, $b_\varepsilon \in [0,\infty)$, $\beta_\varepsilon \in (0,1]$ such that:

$$\inf_{\theta\in\Theta_\varepsilon} P_\theta(x,A) \geq \beta_\varepsilon \nu(A) \mathbf{1}_C(x),\ x \in \mathcal{X}, A \in \mathcal{B}; \tag{8}$$

and

$$\sup_{\theta\in\Theta_\varepsilon} P_\theta V(x) \leq \lambda_\varepsilon V(x) + b_\varepsilon \mathbf{1}_C(x),\ x \in \mathcal{X}. \tag{9}$$

The inequality (9) of (A1) is the so-called drift condition and (8) is the so-called $(1,\varepsilon,\nu)$-minorization condition.

Next, we assume that P_θ is Lipschitz as a function of θ. But first, we need some notations. For any function $f : \mathcal{X} \to \mathbb{R}$ and $W : \mathcal{X} \to [1,\infty)$ we denote $|f|_W := \sup_{x\in\mathcal{X}} \frac{|f(x)|}{W(x)}$, and introduce the set of W-bounded functions $L_W := \{f \text{ meas.},\ f : \mathcal{X} \to \mathbb{R}, |f|_W < \infty\}$. A transition kernel on $(\mathcal{X},\mathcal{B})$ operates on measurable real-valued functions f as $Pf(x) = \int P(x,dy)f(y)$ and the product of two transition kernels P_1 and P_2 is the transition kernel defined as $P_1P_2(x,A) = \int P_1(x,dy)P_2(y,A)$. For two transition kernels P_1 and P_2 we define $\|P_1 - P_2\|_W$, the W-distance between P_1 and P_2 as

$$\|P_1 - P_2\|_W := \sup_{|f|_W\leq 1} |P_1 f - P_2 f|_W\ .$$

(**A2**) For all $\alpha \in [0,1]$, for all $\varepsilon \in (0,\varepsilon_*)$, there exists $K = K(\alpha,\varepsilon) < \infty$ such that for all $\theta,\theta' \in \Theta_\varepsilon$

$$\|P_\theta - P_{\theta'}\|_{V^\alpha} \leq K\,|\theta - \theta'|\,, \tag{10}$$

where V is defined in (A1).

Finally we assume that the step-size sequence is well-behaved:

(A3) $\{\gamma_n\}$ is nonincreasing, $\gamma_n > 0$, $\sum \gamma_n = \infty$, $\sum \gamma^2_{\lfloor n/a \rfloor} < \infty$, and $\sum \gamma_{\lfloor n/a \rfloor}/\lfloor n/a \rfloor < \infty$ for all $a > 0$, where $\lfloor x \rfloor$ denotes the largest integer small or equal to x.

For $B > 0$, we introduce the following stopping time:

$$\tau(B) := \inf\{k \geq 0 : \max_{i,j} k \left(v_k(i) - v_k(j)\right) > B\}, \tag{11}$$

with the usual convention that $\inf \emptyset = \infty$. $\tau(B)$ is the first time where one component will accumulate more than B visits than some other component. Thus the stability of the algorithm means precisely that $\tau(B) = \infty$ for some B. With this in mind, the next results says essentially that if the algorithm is *stable* and (A1-3) hold then it is ergodic.

Theorem 5.1. *Assume (A1-3). Let $B > 0$ be given. Then:*

(i)

$$|\theta_n - \theta^*| \, \mathbf{1}_{\{\tau(B) > n\}} \to 0, \; a.s. \; as \; n \to \infty. \tag{12}$$

(ii) *For any function $f \in L_{V^{1/2}}$, denoting $\bar{f} = f - \pi^*(f)$, we have:*

$$\frac{1}{n} \sum_{k=1}^{n} \bar{f}(X_k, I_k) \mathbf{1}_{\{\tau(B) > k-1\}} \to 0, \; a.s. \; as \; n \to \infty. \tag{13}$$

Proof. See Section 7.1 □

Remark 5.1. 1. Theorem 5.1 asserts that under (A1-3), the WL algorithm will converge to the right limit on *stable* paths. This opens the question of checking the stability of the algorithm. We address this question in some simple cases in Theorem 5.2 below.

2. We discuss (A1-2) in the next Section. The choice of $\{\gamma_n\}$ is important in practice. In Wang and Landau (2001), the sequence $\gamma_n = \gamma_0/2^n$ is used. This works very well in practice but seems to lead to a systematic (but very small) bias. This was also pointed out by Belardinelli and Pereyra (2007). For stochastic approximation, the condition $\sum \gamma_n = \infty$ is in fact necessary. So we suggest taking $\gamma_n = \gamma_0/2^n$ until $\gamma_n < \gamma_*$ for some small threshold γ_* (for example 0.0001), at which point, we take $\gamma_n \propto 1/n$.

5.1.1 Conditions (A1-2) in the case of Metropolis-Hastings kernels

Often (A1-2) do not entail much. Consider the case where each P_θ is a Metropolis-Hastings kernel with invariant distribution π_θ and proposal kernel Q. We give some verifiable conditions which imply (A1-2). Often Q is of the form

$$Q(x,i;A\times\{j\}) = \omega_j(x,i)Q^{(ij)}(x,A), \tag{14}$$

where $\omega_j(x,i)\in[0,1]$ and satisfy $\sum_{j=1}^d \omega_j(x,i)\leq 1$ and $Q^{(i,j)}(x,\cdot)$ is a probability measure on $(\mathcal{X}_j,\mathcal{B}_j,\lambda_j)$ for each $x\in\mathcal{X}_i$. In words, given (x,i), we choose to move to $\mathcal{X}_j\times\{j\}$ with probability $\omega_j(x,i)$ and propose the new state in $\mathcal{X}\times\{j\}$ using $Q^{(ij)}(x,\cdot)$. We refer the reader to Green (1995) for the details in the case of the reversible jump. Tierney (1998) gives a general framework to discuss Metropolis-Hastings kernels that we will follow here for notational simplicity. It includes the reversible jump as a special case.

Let π be the probability measure π_θ when $\theta=(1,\ldots,1)$. Let $\Delta\subseteq\mathcal{X}\times\mathcal{X}$ be a measurable set such that the two probability measures $\pi(dx,i)Q(x,i;dy,j)$ and $\pi(dy,j)Q(y,j;dx,i)$ are equivalent on Δ and mutually exclusive on Δ^c. See Tierney (1998) for the existence of Δ. Define $r(x,i;y,j)=\frac{\pi(dx,i)Q(x,i;dy,j)}{\pi(dy,j)Q(y,j;dx,i)}$, the Radon-Nikodym density of $\pi(dx,i)Q(x,i;dy,j)$ with respect to $\pi(dy,j)Q(y,j;dx,i)$ if $(x,i;y,j)\in\Delta$ and $r(x,i;y,j)=\infty$ otherwise. For $f:\ \mathcal{X}\to\mathbb{R}$ measurable, the Metropolis-Hastings kernel with proposal kernel Q and invariant distribution π_θ operates on f as:

$$P_\theta f(x,i) = M_\theta f(x,i) + r_\theta(x,i)f(x,i),$$

where

$$M_\theta f(x,i) = \sum_{j=1}^d \int_{\mathcal{X}_j} \min\left(1, \frac{\theta(i)}{\theta(j)} r(y,j;x,i)\right) f(y,j)Q(x,i;dy,j),$$

and

$$r_\theta(x,i) = 1 - \sum_{j=1}^d \int_{\mathcal{X}_j} \min\left(1, \frac{\theta(i)}{\theta(j)} r(y,j;x,i)\right) Q(x,i;dy,j).$$

We will write P and M when $\theta=(1,\ldots,1)$. The first result states roughly that if P is geometrically ergodic then (A1-2) hold.

Proposition 5.1. *Suppose that existence of a measurable function* $V: \mathcal{X} \longrightarrow [1,\infty)$, *a set* $C \subset \mathcal{X}$, *a probability measure* ν *on* $(\mathcal{X},\mathcal{B})$ *such that* $\nu(C) > 0$; *constants* $\lambda \in (0,1)$, $b \in [0,\infty)$, $\beta \in (0,1]$ *such that:*

$$M(x,A) \geq \beta\nu(A)\mathbf{1}_C(x),\; x \in \mathcal{X}, A \in \mathcal{B};$$

and

$$PV(x) \leq \lambda V(x) + b\mathbf{1}_C(x),\; x \in \mathcal{X}.$$

Then (A1-2) hold.

Proof. See Section 7.2. □

5.1.2 Geometric ergodicity of Metropolis-Hastings kernels

Proposition 5.1 asserts that if P (defined as P_θ with $\theta(i) \equiv 1$) is geometrically ergodic then (A1-2) hold. Very few results exist in the literature that give verifiable conditions for the geometric ergodicity of the abstract Metropolis-Hastings described above to hold. The next proposition partially fills that gap. It is based on a bound on the *Poincaré constant* for *decomposable Markov chains* developed by Jerrum et al. (2004). We start by introducing the restriction of P on each component $\mathcal{X}_i$. For $i \in \{1,\ldots,d\}$, define

$$P^{[i]}(x,A) = P(x,i;A \times \{i\}) + (1 - P(x,i;\mathcal{X}_i \times \{i\}))\,\mathbf{1}_A(x), \quad x \in \mathcal{X}_i, A \in \mathcal{B}_i. \tag{15}$$

Often each $P^{[i]}$ will be a fixed-dimensional Metropolis-Hastings kernel with target distribution $\pi(dx,i)\mathbf{1}_{\mathcal{X}_i}(x)$. For example if we choose Q as in (14) and $Q^{(ii)}(x,dy) = q^{(ii)}(x,y)\lambda_i(dy)$ then for $x \in \mathcal{X}_i$, $A \in \mathcal{B}_i$

$$P^{[i]}(x,A) = \int_A \min\left(1, \frac{\pi_i(y)}{\pi_i(x)}\frac{\omega_i(y,i)q^{(ii)}(y,x)}{\omega_i(x,i)q^{(ii)}(x,y)}\right)\omega_i(x,i)q^{(ii)}(x,y)\lambda_i(dy) + (1 - r_i(x))\mathbf{1}_A(x).$$

Next we introduce the *projected chain*

$$\bar{P}(i,j) = \frac{1}{\theta^*(i)}\int_{\mathcal{X}_i}\pi(dx,i)\int_{\mathcal{X}_j}\min\left(1, r(y,j;x,i)\right)Q(x,i;dy,j), \quad i,j \in \{1,\ldots,d\}, \tag{16}$$

which approximate the jumps of P from one component to the other. More precisely, $\bar{P}(i,j)$ is the probability of making a transition from i to j in stationarity. By the reversibility of P, $\bar{P}$ is a transition matrix which is also reversible with respect to $(\theta^*(i))$. It is shown in Jerrum et al. (2004) that if $\bar{P}$ is irreducible and aperiodic and if each $P^{[i]}$ has a spectral gap then P has a spectral gap and they give a lower bound on the spectral gap of P. The next proposition is a paraphrase of this result applied to our Metropolis-Hastings kernel P and we omit the proof.

Proposition 5.2. *Suppose that:*

(i) *$\bar{P}$ is irreducible and aperiodic.*

(ii) *for $i \in \{1, \dots, d\}$, $P^{[i]}$ is geometrically ergodic.*

(iii) *there is $\varepsilon > 0$ such that $P(x, i; \{x\} \times \{i\}) \geq \varepsilon$ for all $(x, i) \in \mathcal{X}$.*

Then P is geometrically ergodic.

Remark 5.2. The significance of this proposition is that in trans-dimensional MCMC, this result relates the geometric ergodicity of the trans-dimensional kernel to the geometric ergodicity of the fixed-dimensional kernels on which many results exist in the literature. We impose the additional condition in (iii) that $P(x, i; \{x\} \times \{i\}) \geq \varepsilon$ for some $\varepsilon > 0$ to avoid periodic chains so that the existence of a spectral gap is equivalent to geometric ergodicity. For example if we choose Q as in (14) and let $\sum_{j=1}^{d} \omega_j(x, i) \leq 1 - \varepsilon$ for all $(x, i) \in \mathcal{X}$, then (iii) hold.

5.2 Stability

The stability condition is very diffcult to check in general. But heuristically, since $\theta_n(i)$ increases as $\mathcal{X}_i$ is visited, and the sampling distribution is proportional to $1/\theta_n(i)$, we expect the algorithm to visit all the $\mathcal{X}_i$ often. But we will impose much stronger conditions in order to show that no component receives infinitely more visits than any other component. For two components $i, j \in \{1. \dots, d\}$, we say that i leads to j if there exist $\varepsilon = \varepsilon(i, j) \in (0, \varepsilon_*)$ (where $\varepsilon^* = \min_i \theta^*(i)$) and $K = K(i, j, \varepsilon) < \infty$ such that whenever $\theta(i)/\theta(j) > K$, $P_\theta(x, \mathcal{X}_j \times \{j\}) \geq \varepsilon$ for all $x \in \mathcal{X}_i \times \{i\}$. And we say that j is reachable from i and write $i \rightsquigarrow j$ if there exists a finite path $i_0, i_1, \dots, i_m$ in $\{1 \dots, d\}$, $m \leq d$, such that $i_0 = i$, $i_m = j$ and i_l leads to i_{l+1} for all $l = 0, \dots, m-1$.

Theorem 5.2. *Suppose that $i \rightsquigarrow j$ for all $i, j \in \{1, \dots, d\}$. Then the WL algorithm is stable.*

Proof. See Section 7.3. □

Here is an interesting application. Consider the multicanonical sampling of Section 4.1. Suppose that P_θ is the *Independence-Sampler* with proposal distribution $Q(dx) = q(x)\lambda(dx)$ and invariant distribution $\pi_\theta(dx) \propto \sum_{i=1}^d (h(x)/\theta(i))\mathbf{1}_{\mathcal{X}_i}(x)\lambda(dx)$. Assume that the function $\omega(x) \propto h(x)/q(x)$ is bounded with supremum ω_0. Then clearly, for all $x \in \mathcal{X}_i$,

$$\begin{aligned} P_\theta(x, \mathcal{X}_j) &\geq \min\left(1, \frac{\theta(i)}{\theta(j)}\right) \int_{\mathcal{X}_j} \min\left(1, \frac{\omega(y)}{\omega_0}\right) Q(dy) \\ &\geq \varepsilon_j, \end{aligned}$$

as soon as $\theta(i) \geq \theta(j)$, taking $\varepsilon_j = \int_{\mathcal{X}_j} \min\left(1, \frac{\omega(y)}{\omega_0}\right) Q(dy) > 0$ (since $\pi(\mathcal{X}_i) > 0$). Thus for any $i, j \in \{1, \ldots, d\}$, $i \rightsquigarrow j$ and by Theorem 5.2, the WL algorithm is *stable*. Now, since ω is bounded, each P_θ satisfies a drift condition and a minorization condition which implies (A1-2) by Proposition 5.1. We have just proved the following:

Corollary 5.1. *In the case of multicanonical sampling of Section 4.1, assume that P_θ is the Independence-Sampler with proposal distribution $Q(dx) = q(x)\lambda(dx)$. Assume also that $\omega \propto h/q$ is bounded. Then the WL algorithm is stable and under (A3), we have:*

$$|\theta_n - \theta^*| \to 0; \quad \textit{and} \quad \frac{1}{n}\sum_{k=1}^n f(X_k, I_k) \to \pi^*(f) \;\; \textit{a.s.} \;\; \textit{as } n \to \infty.$$

6 Discussion and open problems

In this paper, we propose an extension of the WL algorithm to general states spaces. The WL algorithm is an efficient algorithm with increasing popularity in the statistical physics community. We show theoretically and through an example that the algorithm can be used effectively to improve on trans-dimensional MCMC algorithms. Another contribution of the paper is that it provides verifiable conditions under which the convergence of the algorithm is guaranteed.

In the trans-dimensional MCMC context, the WL strategy has a number of limitations. One criticism of the approach is that targeting a uniform distribution over all models is waisteful, particularly if the main goal is to estimate the parameters of the best model. But if the main goal of the modeling is in comparing and/or averaging a large number of competing models, this strategy is certainly interesting. Beyong these context-specific arguments, we have made the point in this paper that that under certain conditions, reweighting actually improves on the mixing of trans-dimensional MCMC algorithms.

An interesting and puzzling theoretical question that we leave open is understanding the transcient behavior of the algorithm. Our simulations and many other simulation studies have showed a sharp superior performances of the WL algorithm when the step-size sequence $\{\gamma_n\}$ of the algorithm is controlled as in Wang and Landau (2001) (with the small correction proposed above). Clearly, our analysis in this paper does not offer any explanation to this phenomenon.

7 Proofs

The techniques used here can be found in various form in Benveniste et al. (1990), Delyon (1996), Andrieu et al. (2005). Throughout, C_ε denotes a generic constant whose value can be different from one equation to the next. The key result in the proof of Theorem 5.1 is Lemma 7.2 which states that the weighted sum of the noise process in the stochastic approximation followed by $\{\theta_n\}$ is summable.

7.1 Proof of Theorem 5.1

We start with some preliminary remarks. For any $\varepsilon \in (0, \varepsilon_*)$ and $\alpha \in (0, 1]$, it is known from Markov chains theory that (A1) implies the existence of $C_{\varepsilon,\alpha} < \infty$, $\rho_{\varepsilon,\alpha} \in (0, 1)$ and b_ε as in (A1) such that:

$$\sup_{\theta\in\Theta_\varepsilon} \|P_\theta^n - \pi_\theta\|_{V^\alpha} \leq C_{\varepsilon,\alpha}\rho_{\varepsilon,\alpha}^n, \tag{17}$$

and

$$\sup_{\theta\in\Theta_\varepsilon} \pi_\theta(V) \leq b_\varepsilon. \tag{18}$$

For a proof, see e.g. Baxendale (2005) and the references therein. Define $\xi(\varepsilon) = \inf\{k \geq 0 : \theta_k \notin \Theta_\varepsilon\}$. An easy calculation using (A1) gives that

$$\begin{aligned}\mathbb{E}\left[V(X_n, I_n)\mathbf{1}_{\{\xi(\varepsilon)>n\}}\right] &= \mathbb{E}\left[V(X_n, I_n)\mathbf{1}_{\Theta_\varepsilon}(\theta_0)\cdots\mathbf{1}_{\Theta_\varepsilon}(\theta_n)\right] \\ &\leq \lambda_\varepsilon^n V(X_0, I_0) + b_\varepsilon/(1-\lambda_\varepsilon)\end{aligned}$$

from which we deduce:

$$\sup_n \mathbb{E}\left(V(X_n, I_n)\mathbf{1}_{\{\xi(\varepsilon)>n\}}\right) < \infty. \tag{19}$$

The proof of the theorem will be based on some nice properties of solutions of so-called Poisson equation. These solutions will allow us to obtain a martingale approximation to the process $\sum_{k=1}^n f(X_k, I_k)$. For $f \in L_{V^\alpha}$, $\alpha \in (0,1]$ and $\theta \in \Theta$, define the function

$$h_\theta = \sum_{k=0}^{\infty} P_\theta^k \left(f - \pi_\theta(f)\right). \tag{20}$$

h_θ solves the Poisson equation $f - \pi_\theta(f) = h_\theta - P_\theta h_\theta$. By (A1), h_θ exists and $h_\theta \in L_{V^\alpha}$. For $f \in L_{V^\alpha}$ and $\theta, \theta' \in \Theta_\varepsilon$, denoting $\bar{f}_\theta = f - \pi_\theta(f)$, we have

$$\begin{aligned}\left|\pi_\theta(f) - \pi_{\theta'}(f)\right| &= \left|\pi_\theta\left[P_\theta^k \bar{f}_{\theta'}\right]\right| \\ &= \left|\pi_\theta\left[P_{\theta'}^k(\bar{f}_{\theta'})\right] + \sum_{j=1}^k \pi_\theta\left[P_\theta^{k-j}\left(P_\theta - P_{\theta'}\right)P_{\theta'}^{j-1}(\bar{f}_{\theta'})\right]\right| \\ &\leq C_\varepsilon\left(\rho_\varepsilon^k + |\theta - \theta'|\sum_{j=1}^k \rho_\varepsilon^{j-1}\right),\end{aligned}$$

using (A1-2) from which we deduce that there exists a finite constant C_ε such that:

$$\sup_{f \in L_{V^\alpha}} \left|\pi_\theta(f) - \pi_{\theta'}(f)\right| \leq C_\varepsilon \left|\theta - \theta'\right|. \tag{21}$$

The constant C_ε is not necessarily the same from one equation to the other. Similarly, denoting $\bar{P}_\theta$ the operator $P_\theta - \pi_\theta$, we have

$$\begin{aligned}\left|P_\theta^k \bar{f}_\theta - P_{\theta'}^k \bar{f}_{\theta'}\right| &= \left|\bar{P}_\theta^k f - \bar{P}_{\theta'}^k f\right| \\ &= \left|\sum_{j=1}^k \bar{P}_\theta^{k-j}\left(P_\theta - P_{\theta'}\right)\bar{P}_{\theta'}^{j-1} f\right| \\ &\leq C_\varepsilon \left|\theta - \theta'\right| k\rho_\varepsilon^{k-1} V^\alpha,\end{aligned}$$

using (A1-2). This, together with (21) imply the existence of a finite constant C_ε such that for all $\alpha \in (0,1]$, $\theta, \theta' \in \Theta_\varepsilon$:

$$\left|h_\theta - h_{\theta'}\right|_{V^\alpha} + \left|P_\theta h_\theta - P_{\theta'} h_{\theta'}\right|_{V^\alpha} \leq C_\varepsilon \left|\theta - \theta'\right|. \tag{22}$$

In our analysis, we mainly see $\{\theta_n\}$ as a stochastic approximation sequence. The recursion on $\{\theta_n\}$ writes:

$$\begin{aligned} \theta_{n+1}(i) &= \frac{\phi_{n+1}(i)}{\sum_{e=1}^d \phi_{n+1}(e)} \\ &= \frac{\phi_n(i) + \gamma_{a_n}\phi_n(i)\mathbf{1}_{\{I_{n+1}=i\}}}{\sum_{e=1}^d \phi_n(e) + \gamma_{a_n}\phi_n(I_{n+1})} \\ &= \theta_n(i)\frac{1+\gamma_{a_n}\mathbf{1}_{\{I_{n+1}=i\}}}{1+\gamma_{a_n}\theta_n(I_{n+1})} \\ &= \theta_n(i) + \gamma_{a_n} H_i(\theta_n, I_{n+1}) + \gamma_{a_n}^2 r_{i,n}(\theta_n, I_{n+1}), \end{aligned} \tag{23}$$

where $H_i(\theta, I) = \theta(i)\left(\mathbf{1}_{\{I=i\}} - \theta(I)\right)$ and $r_{i,n}(\theta, I) = -\theta(i)\theta(I)\frac{\mathbf{1}_{\{I=i\}}-\theta(I)}{1+\gamma_{a_n}\theta(I)}$.

The mean field function $h_i(\theta) = \pi_\theta(H_i)$ is

$$h_i(\theta) = \frac{\theta^*(i) - \theta(i)}{\sum_{j=1}^d \frac{\theta^*(j)}{\theta(j)}}. \tag{24}$$

Lemma 7.1. *Assume (A3). Let $B > 0$ be given and $c_0 = 2Bd/c$. Then $a_n \geq \lfloor n/c_0 \rfloor$ and*

$$\sum \gamma_{a_n} = \infty, \quad \text{and} \quad \sum \gamma_{a_n}^2 \mathbf{1}_{\{\tau(B)>n\}} \leq \sum \gamma_{\lfloor n/c_0 \rfloor}^2 < \infty, \ a.s.$$

Proof. Since $\{\gamma_n\}$ is nonincreasing and $a_n \leq n$, $\sum \gamma_{a_n} \geq \sum \gamma_n = \infty$. On $\{\tau(B) > n\}$, $\kappa_l - \kappa_{l-1} \leq c_0$ for all $l \geq 1$. Therefore $a_n \geq \lfloor n/c_0 \rfloor$ and $\sum \gamma_{a_n}^2 \mathbf{1}_{\{\tau(B)>n\}} \leq \sum \gamma_{\lfloor n/c_0 \rfloor}^2 < \infty$. □

Lemma 7.2. *Assume (A1-3). Let $B > 0$ be given. Let $\{\gamma'_n\}$ be a sequence that satisfies (A3) and such that $\sum \gamma_{\lfloor n/a \rfloor}\gamma'_{\lfloor n/a \rfloor} < \infty$ for all $a > 0$. For $\theta \in \Theta$, Let $H_\theta : \mathcal{X} \to \mathbb{R}$ be a measurable function such that $H_\theta \in L_{V^{1/2}}$. Then:*

$$\sum_{k=0}^{\infty} \gamma'_{a_k} \mathbf{1}_{\{\tau(B)>k\}} \left[H_{\theta_k}(X_{k+1}, I_{k+1}) - \pi_{\theta_k}(H_{\theta_k})\right] < \infty, \ \ a.s. \tag{25}$$

Proof. Clearly: for $B > 0$, there exists $\varepsilon \in (0, \varepsilon^*)$ such that for all n, $\tau(B) > n$ implies $\xi(\varepsilon) := \inf\{k \geq 0 : \ \theta_k \notin \Theta_\varepsilon\} > n$. From (A1), $h_\theta \in L_{V^{1/2}}$ exists that solves the Poisson equation $h_\theta - P_\theta h_\theta = H_\theta - \pi_\theta(H_\theta)$ for all $\theta \in \Theta_\varepsilon$. Using this we can write:

$$\sum_{k=0}^{n} \gamma'_{a_k} \mathbf{1}_{\{\tau(B)>k\}} \left[H_{\theta_k}(X_{k+1}, I_{k+1}) - \pi_{\theta_k}(H_{\theta_k})\right] = \sum_{k=0}^{n} \gamma'_{a_k} \mathbf{1}_{\{\tau(B)>k\}} \left(U^{(1)}_{k+1} + U^{(2)}_{k+1} + U^{(3)}_{k+1}\right),$$

where

$$U^{(1)}_{k+1} = h_{\theta_k}(X_{k+1}, I_{k+1}) - P_{\theta_k} h_{\theta_k}(X_k, I_k),$$

$$U^{(2)}_{k+1} = P_{\theta_k} h_{\theta_k}(X_k, I_k) - P_{\theta_{k+1}} h_{\theta_{k+1}}(X_{k+1}, I_{k+1}),$$

$$U^{(3)}_{k+1} = P_{\theta_{k+1}} h_{\theta_{k+1}}(X_{k+1}, I_{k+1}) - P_{\theta_k} h_{\theta_k}(X_{k+1}, I_{k+1}).$$

Clearly, $M_n = \sum_{k=0}^{n} \gamma'_{a_k} \mathbf{1}_{\{\tau(B)>k\}} U^{(1)}_{k+1}$ is a martingale and using Lemma 7.1, (19) and since h_θ and $P_\theta h_\theta \in L_{V^{1/2}}$, we have:

$$\begin{aligned}
\mathbb{E}\left(M_n^2\right) &\leq C_\varepsilon \sum_{k=0}^{n} \mathbb{E}\left[(\gamma'_{a_k})^2 \mathbf{1}_{\{\tau(B)>k\}} V(X_{k+1}, I_{k+1})\right] \\
&\leq C_\varepsilon \sum_{k=0}^{n} (\gamma'_{\lfloor k/c_0 \rfloor})^2 \mathbb{E}\left[\mathbf{1}_{\{\tau(B)>k\}} V(X_{k+1}, I_{k+1})\right] \\
&\leq C_\varepsilon \sum_{k=0}^{\infty} (\gamma'_{\lfloor k/c_0 \rfloor})^2 < \infty.
\end{aligned}$$

By Doob's convergence theorem for martingales, $\sum_{k=0}^{\infty} \gamma'_{a_k} \mathbf{1}_{\{\tau(B)>k\}} U^{(1)}_{k+1}$ is finite a.s.

On $\{\tau(B) = l\}$, $l < \infty$, $\sum_{k=0}^{\infty} \gamma'_{a_k} \mathbf{1}_{\{\tau(B)>k\}} U^{(2)}_{k+1} = \sum_{k=0}^{l-1} \gamma'_{a_k} U^{(2)}_{k+1}$ which is finite almost surely. On $\{\tau(B) = \infty\}$, we can write:

$$\begin{aligned}
\mathbf{1}_{\{\tau(B)=\infty\}} \sum_{k=0}^{n} \gamma'_{a_k} \mathbf{1}_{\{\tau(B)>k\}} U^{(2)}_{k+1} &= \gamma'_{a_0} P_{\theta_0} h_{\theta_0}(X_0, I_0) - \gamma'_{a_n} \mathbf{1}_{\{\tau(B)=\infty\}} P_{\theta_{n+1}} h_{\theta_{n+1}}(X_{n+1}, I_{n+1}) \\
&\quad + \sum_{k=0}^{n-1} \left(\gamma'_{a_{k+1}} - \gamma'_{a_k}\right) \mathbf{1}_{\{\tau(B)=\infty\}} P_{\theta_{k+1}} h_{\theta_{k+1}}(X_{k+1}, I_{k+1}).
\end{aligned}$$

$\mathbb{E}\left[\left(\gamma'_{a_n} \mathbf{1}_{\{\tau(B)=\infty\}} P_{\theta_{n+1}} h_{\theta_{n+1}}(X_{n+1}, I_{n+1})\right)^2\right] \leq C_\varepsilon (\gamma'_{\lfloor n/c_0 \rfloor})^2$ and $\sum (\gamma'_{\lfloor n/c_0 \rfloor})^2 < \infty$ thus $\gamma'_{a_n} \mathbf{1}_{\{\tau(B)=\infty\}} P_{\theta_{n+1}} h_{\theta_{n+1}}(X_{n+1}, I_{n+1})$ converges a.s. to 0.

$$\sum_{k=0}^{n-1} \left| \left(\gamma'_{a_{k+1}} - \gamma'_{a_k}\right) \mathbf{1}_{\{\tau(B)=\infty\}} P_{\theta_{k+1}} h_{\theta_{k+1}}(X_{k+1}, I_{k+1}) \right| \leq C_\varepsilon \sum_{k=0}^{\infty} \left(\gamma'_{a_k} - \gamma'_{a_{k+1}}\right) \mathbf{1}_{\{\tau(B)=\infty\}} V^{1/2}(X_{k+1}, I_{k+1}).$$

Since a_k only changes at the stopping times κ_i, we have

$$\begin{aligned}
\mathbb{E}\left[\sum_{k=0}^{\infty}\left(\gamma'_{a_k}-\gamma'_{a_{k+1}}\right)\mathbf{1}_{\{\tau(B)=\infty\}}V^{1/2}(X_{k+1},I_{k+1})\right] &= \mathbb{E}\left[\sum_{k=1}^{\infty}\left(\gamma'_{k-1}-\gamma'_{k}\right)\mathbf{1}_{\{\tau(B)=\infty\}}V^{1/2}(X_{\kappa_k+1},I_{\kappa_k+1})\right]\\
&= \sum_{k=1}^{\infty}\left(\gamma'_{k-1}-\gamma'_{k}\right)\mathbb{E}\left[\mathbf{1}_{\{\tau(B)=\infty\}}V^{1/2}(X_{\kappa_k+1},I_{\kappa_k+1})\right]\\
&\leq C_\varepsilon\sum_{k=1}^{\infty}\left(\gamma'_{k-1}-\gamma'_{k}\right)\\
&\leq C_\varepsilon\gamma'_0.
\end{aligned}$$

By Lebesgue's dominated convergence theorem, we can conclude that

$$\mathbb{E}\left[\left|\sum_{k=0}^{\infty}\left(\gamma'_{a_{k+1}}-\gamma'_{a_k}\right)\mathbf{1}_{\{\tau(B)=\infty\}}P_{\theta_{k+1}}h_{\theta_{k+1}}(X_{k+1},I_{k+1})\right|\right]<\infty,$$

This is sufficient to conclude that $\sum_{k=0}^{\infty}\gamma'_{a_k}\mathbf{1}_{\{\tau(B)>k\}}U^{(2)}_{k+1}$ is finite almost surely.

Using (22):

$$\begin{aligned}
\sum_{k=0}^{n}\gamma'_{a_k}\mathbf{1}_{\{\tau(B)>k\}}\left|U^{(3)}_{k+1}\right| &\leq C_\varepsilon\sum_{k=0}^{\infty}\gamma'_{a_k}\gamma_{a_k}\mathbf{1}_{\{\tau(B)>k\}}V^{1/2}(X_{k+1},I_{k+1})\\
&\leq C_\varepsilon\sum_{k=0}^{\infty}\gamma'_{\lfloor n/c_0\rfloor}\gamma_{\lfloor n/c_0\rfloor}\mathbf{1}_{\{\tau(B)>k\}}V^{1/2}(X_{k+1},I_{k+1})
\end{aligned}$$

and

$$E\left[\sum_{k=0}^{\infty}\gamma'_{\lfloor n/c_0\rfloor}\gamma_{\lfloor n/c_0\rfloor}\mathbf{1}_{\{\tau(B)>k\}}V^{1/2}(X_{k+1},I_{k+1})\right]\leq C_\varepsilon\sum_{k=0}^{\infty}\gamma'_{\lfloor n/c_0\rfloor}\gamma_{\lfloor n/c_0\rfloor}<\infty$$

by (19). With Lebesgue's dominated convergence theorem, we deduce that

$$\mathbb{E}\left[\sum_{k=0}^{\infty}\gamma'_{a_k}\mathbf{1}_{\{\tau(B)>k\}}\left|U^{(3)}_{k+1}\right|\right]\leq C_\varepsilon\sum_{k=0}^{\infty}\gamma'_{\lfloor n/c_0\rfloor}\gamma_{\lfloor n/c_0\rfloor}<\infty$$

which implies that $\sum_{k=0}^{n}\gamma'_{a_k}\mathbf{1}_{\{\tau(B)>k\}}U^{(3)}_{k+1}$ converges almost surely to a finite limit. This completes the proof of the lemma. □

We are now in position to prove Theorem 5.1. We start with (i).

Proposition 7.1. *Assume (A1-3) and let $B>0$ be given. Then $|\theta_n-\theta^*|\,\mathbf{1}_{\{\tau(B)>n\}}\to 0$ with probability one as $n\to\infty$.*

Proof. The idea of the proof is borrowed from Delyon (1996). We saw in (23) that

$$\theta_{n+1} = \theta_n + \gamma_{a_n} H(\theta_n, I_{n+1}) + \gamma_{a_n}^2 r_n,$$

where $H = (H_1, \ldots, H_d)$, $r_n = (r_{n,1}, \ldots, r_{n,d})$, $H_i(\theta, I) = \theta(i)\left(\mathbf{1}_{\{I=i\}} - \theta(I)\right)$ and $r_{i,n} = -\theta(i)\theta(I)\frac{\mathbf{1}_{\{I=i\}}-\theta(I)}{1+\gamma_{a_n}\theta(I)}$. We note that $|H| \leq 1$ and $|r_n| \leq 1$. Let $\varepsilon \in (0, \varepsilon^*)$ be such that when $\tau(B) > n$, $\xi(\varepsilon) > n$. We recall that the mean field function of the recursion is $h_i(\theta) = (\theta^*(i) - \theta(i))/\sum_{j=1}^d \frac{\theta^*(j)}{\theta(j)}$. We introduce $\theta_n' = \theta_n + \sum_{j=n}^{\infty} \gamma_{a_j} \mathbf{1}_{\{\tau(B)>j\}} \left[H(\theta_j, I_{j+1}) - h(\theta_j)\right]$. From Lemma 7.2, $|\theta_n' - \theta_n| \to 0$ almost surely. $\{\theta_n'\}$ satisfies the recursion

$$\theta_{n+1}' = \theta_n' + \gamma_{a_n} h(\theta_n) + \gamma_{a_n} \mathbf{1}_{\{\tau(B)\leq n\}} \left[H(\theta_n, I_{n+1}) - h(\theta_n)\right] + \gamma_{a_n}^2 r_n.$$

We can then deduce:

$$\begin{aligned}
\left|\theta_{n+1}' - \theta^*\right|^2 \mathbf{1}_{\{\tau(B)>n\}} &= \left|\theta_n' - \theta^*\right|^2 \mathbf{1}_{\{\tau(B)>n\}} + 2\gamma_{a_n} \left\langle \theta_n' - \theta^*, h(\theta_n)\right\rangle \mathbf{1}_{\{\tau(B)>n\}} + \gamma_{a_n} \mathbf{1}_{\{\tau(B)>n\}} r_n' \\
&\leq (1 - 2\varepsilon\gamma_{a_n}) \left|\theta_n' - \theta^*\right|^2 \mathbf{1}_{\{\tau(B)>n\}} + \gamma_{a_n} \mathbf{1}_{\{\tau(B)>n\}} \left(r_n' + \left\langle \theta_n' - \theta^*, h(\theta_n) - h(\theta_n')\right\rangle\right).
\end{aligned}$$

where $r_n' \to 0$ almost surely as $n \to \infty$. Since θ_n remains in the compact Θ_ε, h is continuous and since $|\theta_n' - \theta_n| \to 0$, it follows that $\langle \theta_n' - \theta^*, h(\theta_n) - h(\theta_n')\rangle \to 0$. We can summarize the situation like this. Writting $U_n = \left|\theta_n' - \theta^*\right|^2 \mathbf{1}_{\{\tau(B)>n\}}$, we have:

$$U_{n+1} \leq (1 - 2\varepsilon\gamma_{a_n})U_n + \gamma_{a_n} r_n'', \tag{26}$$

where $r_n'' \to 0$ as $n \to \infty$. This implies that $U_n \to 0$ which, given Lemma 7.2 proves the Proposition.

To see why $U_n \to 0$, let $\delta > 0$ be given. Take $n_0 > 0$ such that for $n \geq n_0$, $\left|r_n''\right| \leq 2\varepsilon\delta$ and $(1 - 2\varepsilon\gamma_{a_n})\mathbf{1}_{\{\tau(B)>n\}} > 0$. Then for $n \geq n_0$, $(U_{n+1} - \delta) \leq (1 - 2\varepsilon\gamma_{a_n})(U_n - \delta) + 2\varepsilon\gamma_{a_n}(r_n''/2\varepsilon - \delta) \leq (1 - 2\varepsilon\gamma_{a_n})(U_n - \delta)$ which implies that $\limsup(U_n - \delta) \leq 0$ and since $\delta > 0$ is arbitrary, we conclude that $\lim U_n = 0$.

□

Proposition 7.2. *Assume (A1-3) and let $B > 0$ be given. For any function $f \in L_{V^{1/2}}$, denoting $\bar{f} = f - \pi^*(f)$, we have:*

$$\frac{1}{n}\sum_{k=1}^{n} \bar{f}(X_k, I_k)\mathbf{1}_{\{\tau(B)>k-1\}} \to 0, \ \textit{a.s. as } n \to \infty. \tag{27}$$

Proof. In view of Proposition 7.1 and (21), we only need to show that

$$\frac{1}{n}\sum_{k=1}^{n}\left(f(X_k, I_k) - \pi_{\theta_{k-1}}(f)\right)\mathbf{1}_{\{\tau(B)>k-1\}} \to 0 \;\; a.s. \tag{28}$$

Kronecker's Lemma applied to (25) of Lemma 7.2 with $\gamma_n' = 1/n$, $H_\theta = f$ yields (28). □

7.2 Proof of Proposition 5.1

Lemma 7.3. *Under the conditions of Proposition 5.1, (A2) hold.*

Proof. Let $\varepsilon \in (0, \varepsilon^*)$ and $\alpha \in (0,1]$. For $\theta \in \Theta_\varepsilon$, $|f| \le V^\alpha$ and $(x,i) \in \mathcal{X}$, we have:

$$P_\theta f(x,i) = M_\theta f(x,i) + f(x,i)\left(1 - M_\theta \mathbf{e}(x,i)\right),$$

where $\mathbf{e}(x,i) \equiv 1$, and

$$M_\theta f(x,i) = \sum_{j=1}^{d}\int_{\mathcal{X}_j} \min\left(1, \frac{\theta(i)}{\theta(j)} r(y,j;x,i)\right) f(y,j) Q(x,i;dy,j).$$

As a consequence, for $\theta_2, \theta_1 \in \Theta_\varepsilon$:

$$\|P_{\theta_2} - P_{\theta_1}\|_{V^\alpha} \le \|M_{\theta_2} - M_{\theta_1}\|_{V^\alpha} + \|M_{\theta_2} - M_{\theta_1}\|_{TV},$$

where $\|P\|_{TV} = \|P\|_V$ with $V \equiv 1$. For $(x,i) \in \mathcal{X}$, the function $\theta \to M_\theta f(x,i)$ is differentiable and:

$$\sum_{j=1}^{d}\left|\frac{\partial}{\partial \theta_j} M_\theta f(x,i)\right| \le C_\varepsilon \sum_{j=1}^{d} M\,|f|\,(x,i).$$

(A2) then follows by the mean-value theorem and the drift condition on P. □

Next we show that for any $\varepsilon \in (0, \varepsilon^*)$, the family $(P_\theta)_{\theta\in\Theta_\varepsilon}$ satisfies a uniform (in θ) drift condition.

Lemma 7.4. *Under the conditions of Proposition 5.1, (A1) hold.*

Proof. The minorization is immediat. Since $\min(1, ab) \ge \min(1,a)\min(1,b)$ for $a, b > 0$, $P_\theta(x,i;A) \ge M_\theta(x,i;A) \ge (1/\varepsilon)M_0(x,i;A) \ge \beta/\varepsilon\nu(A)\mathbf{1}_C(x,i)$.

The argument to show the uniform drift is fairly simple and we only sketch it. Let $B_{\theta,r} = \Theta_\varepsilon \cap B(\theta, r)$ the open ball of Θ_ε with center $\theta \in \Theta_\varepsilon$ and radius $r > 0$. It follows from Lemma 7.3 that by choosing $r > 0$ small enough, if P_θ satisfies a drift condition toward a small set C, then the family $\{P_{\theta'},\ \theta' \in B_{\theta,r}\}$ satisfies a uniform (in θ') drift condition toward C. Therefore, starting from P, we can find an open coverage of Θ_ε by the $B_{\theta,r}$ such that on each such $B_{\theta,r}$ a uniform drift toward C hold. Since Θ_ε is compact it admits a finite coverage by the $B_{\theta,r}$ and using the maximum of the constants of the drift condition toward C for each such ball, we get a uniform drift toward C for $\{P_\theta,\ \ \theta \in \Theta_\varepsilon\}$. □

7.3 Proof of Theorem 5.2

Proof. It will sufficient to show that for $i, j \in \{1, \ldots, d\}$, such that $i \rightsquigarrow j$, $n(v_n(i) - v_n(j))$ is bounded from above, where $v_n(i) = \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{I_k=i\}}$. This clearly implies the stability of the algorithm. To do so, it suffices to show that for $i, j \in \{1, \ldots, d\}$ such that i leads to j:

$$\limsup_{n\to\infty} n(v_n(i) - v_n(j)) < \infty,\ a.s. \tag{29}$$

Indeed, for $i \neq j \in \{1, \ldots, d\}$ arbitrary such that $i \rightsquigarrow j$, there exists a path $i_0, \ldots, i_m$, $m \leq d$, $i_0 = i$, $i_m = j$ and i_l leads to i_{l+1}, $l = 0, \ldots, m-1$. So if (29) hold, $n(v_n(l) - v_n(l+1))$ is bounded from above and so is $n(v_n(i) - v_n(j)) = \sum_{l=0}^{m-1} n(v_n(l) - v_n(l+1))$.

To prove (29), define $\alpha_k = (1 + \gamma_0)^{2k}$ and suppose that $\limsup_{n\to\infty} n(v_n(i) - v_n(j)) = \infty$. This implies the existence of increasing sequence of integers $\{n_k,\ k \geq 1\}$ such that $n_k(v_{n_k}(i) - v_{n_k}(j)) > \alpha_k$ and $(X_{n_k}, I_{n_k}) \in \mathcal{X}_i \times \{i\}$ but $(X_{n_{k+1}}, I_{n_{k+1}}) \notin \mathcal{X}_j \times \{j\}$ for all $k \geq 1$. Clearly, $n_k(v_{n_k}(i) - v_{n_k}(j)) > \alpha_k$ implies that $\theta_{n_k}(i)/\theta_{n_k}(j) > e^{\alpha_k}$. But, since i leads to j, we can find $\varepsilon > 0$ and k_0 such that for $k \geq k_0$:

$$\Pr\left((X_{n_{k+1}}, I_{n_{k+1}}) \notin \mathcal{X}_j \times \{j\} | \mathcal{F}_{n_k}, (X_{n_k}, I_{n_k}) \in \mathcal{X}_i \times \{i\}, \theta_{n_k}(i)/\theta_{n_k}(j) > e^{\alpha_k}\right) \leq (1 - \varepsilon).$$

Thus $\Pr\left(\limsup_{n\to\infty} n(v_n(i) - v_n(j)) = \infty\right) \leq lim_{k\to\infty}(1 - \varepsilon)^k = 0.$ □

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References

Andrieu, C. and Atchade, Y. F. (2005). On the efficiency of adaptive mcmc algorithms. *Technical report, MCMC preprints* .

Andrieu, C. and Moulines, É. (2006). On the ergodicity properties of some adaptive MCMC algorithms. *Ann. Appl. Probab.* **16** 1462–1505.

Andrieu, C., Moulines, É. and Priouret, P. (2005). Stability of stochastic approximation under verifiable conditions. *SIAM J. Control Optim.* **44** 283–312 (electronic).

Andrieu, C. and Robert, C. P. (2001). Controlled mcmc for optimal sampling. *Technical report, Université Paris Dauphine, Ceremade 0125* .

Atchade, Y. F. and Liu, J. S. (2006). Discussion of the paper by kou, zhou and wong. *Annals of Statistics* **To appear**.

Atchade, Y. F. and Rosenthal, J. S. (2005). On adaptive markov chain monte carlo algorithm. *Bernoulli* **11** 815–828.

Baxendale, P. H. (2005). Renewal theory and computable convergence rates for geometrically ergodic markov chains. *Annals of Applied Probability* **15** 700–738.

Belardinelli, R. E. and Pereyra, V. D. (2007). Wang-landau algorithm: a theoretical analysis of the saturation error. *Technical Report, Arxiv.org* .

Benveniste, A., Métivier, M. and Priouret, P. (1990). *Adaptive Algorithms and Stochastic approximations*. Applications of Mathematics, Springer, Paris-New York.

Berg, B. A. and Neuhaus, T. (1992). Multicanonical ensemble: A new approach to simulate first-order phase transitions. *Phys. Rev. Lett.* **68**.

Brockwell, A. E. and Kadane, J. B. (2005). Identification of regeneration times in MCMC simulation, with application to adaptive schemes. *J. Comput. Graph. Statist.* **14** 436–458.

Brooks, S. P., Giudici, P. and Roberts, G. O. (2003). Efficient construction of reversible jump Markov chain Monte Carlo proposal distributions. *J. R. Stat. Soc. Ser. B Stat. Methodol.* **65** 3–55.

Delyon, B. (1996). General results on the convergence of stochastic algorithms. *IEEE Trans. Automat. Control* **41** 1245–1255.

Geyer, C. J. and Thompson, E. (1995). Annealing markov chain monte carlo with applications to pedigree analysis. *Journal of the American Statistical Association* **90** 909–920.

Gilks, W. R., Roberts, G. O. and Sahu, S. K. (1998). Adaptive Markov chain Monte Carlo through regeneration. *J. Amer. Statist. Assoc.* **93** 1045–1054.

Green, P. J. (1995). Reversible jump Markov chain Monte Carlo computation and Bayesian model determination. *Biometrika* **82** 711–732.

Haario, H., Saksman, E. and Tamminen, J. (2001). An adaptive metropolis algorithm. *Bernoulli* **7** 223–242.

Jerrum, M., Son, J.-B., Tetali, P. and Vigoda, E. (2004). Elementary bounds on poincaré and log-sobolev constants for decomposable markov chains. *Annals of Applied Probability* **14** 1741–1765.

LEE, H. K., OKABE, Y. and LANDAU, D. P. (2005). Convergence and refinement of the wang-landau algorithm. *Technical Report* .

LIANG, F., LIU, C. and CARROLL, R. J. (2007). Stochastic approximation in monte carlo computation. *JASA* **102** 305–320.

MARIN, J. M. and MENGERSEN, C. P., K. AD ROBERT (2005). Bayesian modelling and inference on mixtures of distributions. *Handbook of Statistics* **25**.

MARINARI, E. and PARISI, G. (1992). Simulated tempering: A new monte carlo schemes. *Europhysics letters* **19** 451–458.

MIRA, A. and SARGENT, D. J. (2003). A new strategy for speeding Markov chain Monte Carlo algorithms. *Stat. Methods Appl.* **12** 49–60.

RICHARDSON, S. and GREEN, P. J. (1997). On Bayesian analysis of mixtures with an unknown number of components. *J. Roy. Statist. Soc. Ser. B* **59** 731–792.

ROSENTHAL, J. S. and ROBERTS, G. O. (2005). Coupling and ergodicity of adaptive mcmc. *Technical Report, MCMC preprints* .

TIERNEY, L. (1998). A note on Metropolis-Hastings kernels for general state spaces. *Ann. Appl. Probab.* **8** 1–9.

WANG, F. and LANDAU, D. P. (2001). Efficient, multiple-range random walk algorithm to calculate the density of states. *Physical Review Letters* **86** 2050–2053.